CS 101: Fundamentals of Computer Science I

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Introduction

- About me
  - Ph.D. in Computer Science, *Virginia Tech*
  - Research: AI-based Smart Systems, Machine Learning, Explainable AI
- **About you**
  - Your name
  - Your major
  - Have you programmed before? If yes, which language(s)?
  - What do you think programming is?

About This Course

- What will you learn?
  - Think like a programmer
  - Solve problems systematically
  - Write C programs
  - Build a foundation for later CS courses

This course is NOT about memorizing syntax.

It is about learning problem solving.

What is Programming?

- Programming is the process of writing instructions that tell a computer how to solve a problem.

- **Algorithm:** A step-by-step procedure for solving a problem.
- **Program:** An algorithm written in a programming language that a computer can execute.

Example: Odd or Even?

Problem

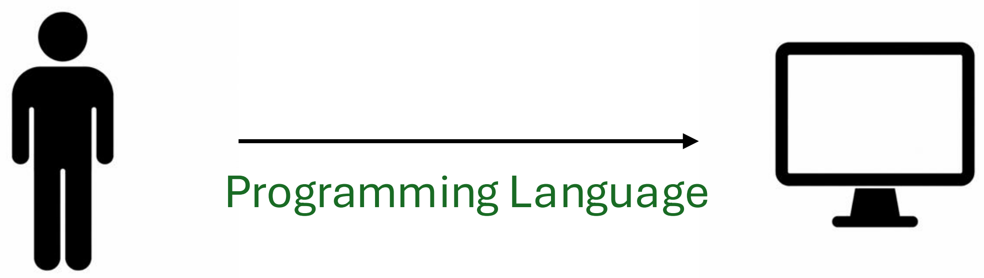
- Determine whether an integer is odd or even.

Algorithm

1. Input an integer **n**.
2. Divide **n** by **2**.
3. Check the remainder.
4. If the remainder is **0**, output **"Even"**.
5. Otherwise, output **"Odd"**.

What Is a Programming Language?

- How do we communicate with a computer?
  - Computers understand only machine instructions.
  - Humans write programs using programming languages.
  - A programming language lets us express an algorithm in a form that can later be translated into machine code.



An algorithm becomes a program when it is written in a programming language.

Why Learn C?

- Why is C still important?
  - Foundation of many modern languages
  - Fast and efficient
  - Close to computer hardware
  - Used in
    - Operating systems
    - Embedded systems
    - Robotics
    - Microcontrollers
    - Device drivers

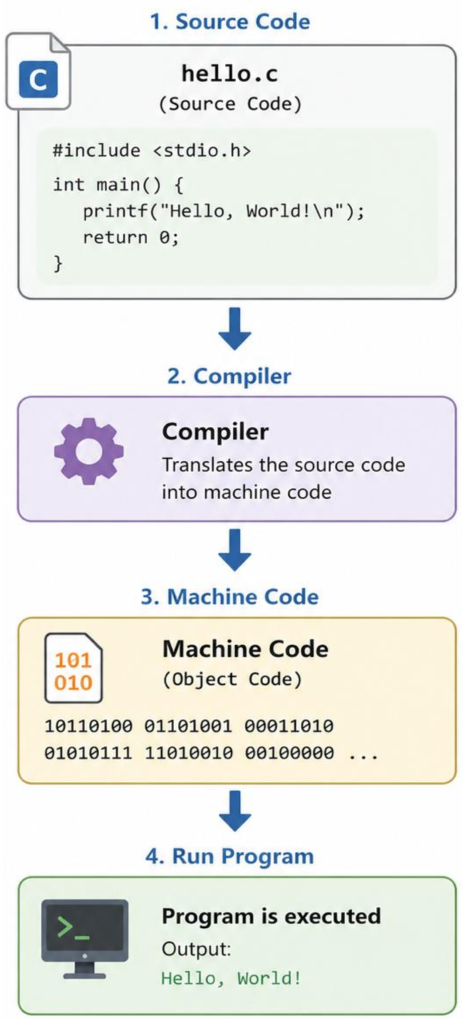
A First C Program

```c
#include <stdio.h> // Gives us access to printf()

int main(){  // Every C program starts here
    printf("Hello World!\n");  // Print a message on the screen
    return 0; // End of the program
}
```

How Does the Computer Understand C?

- Computers understand only machine code (0s and 1s).
- A compiler translates your C program into machine code that the computer can execute.



Key Takeaways

- What programming is
- Algorithm vs. program
- Programming language
- Why C
- First C program